

ADITYA SINGH RATHORE

FPGA & RTL Design · Hardware Verification · Embedded Systems · Xilinx Vivado
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PROFILE

Electrical Engineering undergraduate (NIT Hamirpur, 2nd year) currently on a DRDO-funded research internship at IIT Jodhpur. Have independently designed, integrated, and hardware-verified two complete FPGA SoC systems — a SweRVolf RISC-V SoC with MIG DDR3, and a MicroBlaze system with DDR3 — on the Digilent Arty A7-100T using Vivado 2025.1. Career goal: VLSI verification engineering at tier-1 semiconductor companies. Available for freelance FPGA design, RTL development, verification, and technical documentation work with fast turnaround.

FREELANCE SERVICES — WHAT I DELIVER

RTL Design & Simulation

- Verilog / SystemVerilog modules
- FSM, pipeline, ALU, UART design
- Lint-clean, synthesis-ready RTL
- GTKWave / Vivado simulation

Functional Verification

- Directed & constrained-random TBs
- SystemVerilog Assertions (SVA)
- Functional coverage planning
- Bug reporting & waveform debug

FPGA Implementation

- Vivado block design & IP integration
- Timing closure & XDC constraints
- ILA / Chipscope hardware debug
- Bitstream + test report delivery

AXI / SoC Integration

- AXI4, AXI-Lite, AXI Stream IPs
- MIG DDR3 memory controller
- CDC: 2-flop sync, AXI Clock Conv.
- MicroBlaze / RISC-V SoC bring-up

Embedded C / Vitis

- Bare-metal C on MicroBlaze / ARM
- Peripheral driver development
- Vitis workspace setup & debug
- HW-SW co-verification

UVM & Advanced Verification

- UVM TB: sequencer-driver-monitor
- Scoreboard & coverage collection
- Cocotb Python-based testbenches
- (Actively learning — available soon)

Python & Scripting

- Waveform parsing & report scripts
- Vivado Tcl batch automation
- MATLAB / Simulink modelling
- Regression & result automation

Technical Documentation

- Detailed design reports (LaTeX/Word)
- Block diagrams & architecture docs
- Testbench documentation
- Academic project write-ups

Linux on FPGA / SoC

- PetaLinux / embedded Linux basics
- Device tree & driver fundamentals
- RISC-V toolchain (riscv-gnu)
- Bare-metal to Linux migration

RESEARCH INTERNSHIP EXPERIENCE

FPGA / Hardware Security Research Intern

IIT Jodhpur · May – July 2026

DRDO-funded project · Supervisor: Asst. Prof. Binod Kumar · Arty A7-100T, Vivado 2025.1

Project 1: SweRVolf RISC-V SoC + MIG DDR3 Integration (Original Engineering Contribution)

- **mig_ddr3_top RTL wrapper:** encapsulates Xilinx MIG 7-Series DDR3 controller behind a clean AXI4 slave interface; PLLE2_BASE generates 333.33 MHz system clock and 200 MHz IDELAYCTRL reference internally.
- Four-domain clock architecture (40 / 81.25 / 200 / 333.33 MHz); all CDC via AXI Clock Converter + ASYNC_REG two-flop synchronisers — zero metastability violations.
- Hierarchical reset sequencing: PLL lock → MIG reset → SoC reset; no AXI traffic before DDR3 PHY calibration completes.
- Hardware verified: init_done=1, init_error=0; 16 KiB write/read/compare with zero errors; write 3.477 MB/s, read 2.316 MB/s @ 40 MHz; timing closure WNS = +1.116 ns.
- Produced 29-section technical report covering full RTL, MIG config, PLL params, ILA strategy, and benchmark analysis — accepted by professor.

Project 2: MicroBlaze Soft-Processor + DDR3 (Vivado Block Design + Vitis Software)

- Complete Vivado block design: MicroBlaze + MIG DDR3 + AXI Interconnect + AXI Timer; bare-metal C application deployed in Vitis.
- Debugged circular reset feedback (ui_clk_sync_rst) and AXI Timer errors using ILA hardware capture.
- CPU/DDR clock sweep: timing closure boundary ~116.4 MHz (WNS = +0.140 ns); proved AXI bus — not DDR — is the throughput bottleneck; D-cache gives 6.3× read speedup.

RTL DESIGN & VERIFICATION PROJECTS

32-bit Pipelined ALU

SystemVerilog · Basys 3 · Full Verification

- 3-stage pipeline (Decode → Execute → Writeback) with stall/flush hazard control; constrained-random testbench, SVA assertions, 100% code + functional coverage.
- Synthesised in Vivado; timing closure verified in GTKWave waveform analysis.

Parameterised UART (TX + RX)

Verilog · Loopback Verified

- Configurable baud rate 9600–115200, 7/8-bit data, None/Even/Odd parity; 16× oversampling on RX; self-checking loopback testbench with fault injection confirming parity-error flag.

Traffic Light Controller · 16-bit Adder (RCA vs CLA) · CMOS Inverter Analysis

Verilog / Schematic

- Moore FSM traffic controller with timed state transitions; RCA vs CLA speed/area comparison via GTKWave timing sim; CMOS inverter VTC, noise margins, switching threshold study.

HDLBits — 120+ Problems Solved

Self-Study

- Combinational logic, sequential circuits, shift registers, FSMs, and arithmetic circuits across all difficulty levels.

TECHNICAL SKILLS

HDL: Verilog, SystemVerilog

Verification: SVA, constrained-random TBs, functional coverage, UVM (learning)

FPGA Boards: Arty A7-100T (Artix-7 XC7A100T), Basys 3 (Artix-7)

EDA / Tools: Vivado 2025.1, Vitis 2025.1, GTKWave, Vivado Tcl

IP / Protocols: AXI4, AXI-Lite, AXI Stream, Xilinx MIG DDR3, Clocking Wizard, ILA

Scripting: Python (cocotb, automation), MATLAB/Simulink (learning), Tcl

SoC / CPU: MicroBlaze, SweRVolf RISC-V, bare-metal C, Vitis SDK

Other: Linux on FPGA (learning), LaTeX, technical report writing, Git

EDUCATION

B.Tech — Electrical Engineering

NIT Hamirpur · 2024 – 2028 (expected) · ID: 24BEE004

Coursework: Digital Electronics, Microprocessors & Microcontrollers (8085/8086, ARM, 8051), Neural Networks & Fuzzy Logic, Power Electronics

Research Intern — FPGA / Hardware Security

IIT Jodhpur · May – July 2026 · DRDO-funded

Two complete FPGA SoC projects delivered and hardware-verified; top performer in lab — M.Tech students and paid project associate regularly consult for technical guidance.

CURRENTLY LEARNING — EXPANDING SKILL SET

UVM (Universal Verification Methodology)

Sequencer-driver-monitor architecture, UVM phases, scoreboard design, coverage-driven verification flow.

Cocotb & Python Verification

Python-based testbench automation using cocotb; waveform parsing scripts; Vivado Tcl batch flows for regression.

Linux on FPGA / PetaLinux

Embedded Linux bring-up on Xilinx SoCs, device tree basics, RISC-V GNU toolchain and bare-metal to OS migration.

Available for freelance work · Fast turnaround · Clear communication · Documented deliverables